Target validation paths — pancreatic-recurrent-kras-g12r-m8f3

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

All six biomarkers in this case read as confirmed on the local report, so the workup here hardens the calls rather than gates a fresh diagnosis. The single essential row is orthogonal KRAS G12R confirmation by plasma ctDNA plus a second-platform tissue NGS: pan-RAS and RAS(ON) sponsors re-test the variant on plasma at screening regardless of the local report, and the same draw seeds the baseline VAF used for on-treatment monitoring at cycle 2 and cycle 4. This re-test gates trial entry for daraxonrasib (RMC-6236, NCT05379985), the daraxonrasib + chemo combination platform (NCT06445062), and the zoldonrasib (RMC-7977) + ivonescimab program (NCT07397338). Five high-priority workups run in parallel: a comprehensive germline panel, p16 IHC on the CDKN2A call, CCND3 alteration-class refinement, KRAS co-mutation NGS (SMAD4 / KEAP1 / STK11 / GNAS / ARID1A), and MTAP reflex testing on the 9p21 locus. Six medium-priority workups round out the dossier.

KRAS G12R

Essential: orthogonal NGS plus baseline ctDNA. Pan-RAS / RAS(ON) trial central labs re-test KRAS G12X on plasma at screening; the orthogonal call locks eligibility for daraxonrasib monotherapy (NCT05379985), daraxonrasib + chemo (NCT06445062), and the RMC-7977 + ivonescimab combination (NCT07397338). Reference platforms are Guardant360 CDx, FoundationOne Liquid CDx, Tempus xF+, Natera Signatera, and Caris Assure; the FDA-approved CDx platforms are the safest choice for trial-entry documentation. G12R is biologically distinct from G12C and G12D (RAS-GTP-binding-deficient per Hobbs 2020), so G12C-specific covalent agents are not the relevant class. Turnaround is one to two weeks.

Three high-priority companions ride on the same NGS order. Co-mutation profiling for SMAD4, KEAP1, STK11, GNAS, and ARID1A frames prognosis and weights any ICI-combination decision: SMAD4 loss in PDAC is the strongest single co-alteration linked to wider metastatic spread, and KEAP1 / STK11 predict diminished ICI benefit on KRAS-mutant backgrounds. A germline hereditary-cancer panel (BRCA1, BRCA2, PALB2, ATM, MLH1, MSH2, MSH6, PMS2, EPCAM, CDKN2A, TP53, STK11) is NCCN-recommended for every PDAC patient regardless of family history. A pathogenic BRCA1 / BRCA2 or PALB2 result opens olaparib maintenance via NCT02184195 plus rucaparib as the platinum-induction-gated PARP follow-on. Germline turnaround is three to six weeks; order in parallel with a genetic-counseling referral.

Two medium-priority resistance reads sit alongside: KRAS copy-number annotation (baseline KRAS amplification and acquired wildtype-KRAS amplification are documented escape mechanisms on the pan-RAS class), and serial KRAS G12R ddPCR or tumor-informed MRD at cycle 2 and cycle 4 of any KRAS-directed therapy, which flags VAF rebound weeks ahead of RECIST imaging.

TP53

The local report annotates an inactivating mutation, but the functional class matters: pure loss-of-function (truncations, splice) reads differently from dominant-negative missense hotspots (R175H, R248W, R273H) and from gain-of-function variants, and any TP53-restoration or MDM2-axis strategy gates on the precise call. The workup is a same-day re-read of the existing NGS report against the IARC TP53 database for variant annotation in HGVS notation. Does not gate the KRAS axis; refines any downstream cell-cycle / DDR conversation.

CDKN2A

Two confirmatory reads. p16 IHC (clone E6H4 on archival FFPE, one to two week turnaround) is the orthogonal protein-level check on the NGS copy-number call; CDK4/6-inhibitor trial central labs prefer p16-negative status or biallelic CDKN2A loss for enrollment in CDKN2A-loss-enriched arms. A discordance (NGS-loss but IHC-retained) suggests subclonal or assay-driven calls and tempers expectations for CDK4/6 monotherapy.

Multi-region p16 IHC on a second tumor block (primary plus metastatic site if available) reads PDAC clonal heterogeneity at the next tier. A patchy result does not gate enrollment, but it changes how a CDK4/6-inhibitor response is interpreted and may push the board toward combination strategies.

The 9p21 MTAP reflex (callable from the same NGS panel) is the other operational read: CDKN2A and MTAP are co-deleted in roughly 80 to 90 percent of CDKN2A homozygous deletions, and MTAP co-deletion is the eligibility threshold for the MTA-cooperative PRMT5 class (anvumetostat on NCT06360354, BMS-986504 on NCT07492680). Decisive for any decision on whether the dual-feature daraxonrasib + anvumetostat regimen is on the table.

CCND3

The local NGS report names a CCND3 alteration without specifying class. An amplification or activating mutation supports cyclin-D-axis dependency and CDK4/6 sensitivity; a passenger missense without copy gain carries weaker mechanistic weight. Cyclin-D-degrader and selective CDK4 programs (PF-07220060, fadraciclib, INCB123667) gate on copy-number gain or known activating mutations, so the class call frames whether these axes belong on the page at all. Usually returnable as an annotation re-read of the existing NGS panel; ask the molecular pathologist for the alteration class explicitly. Turnaround is two to three weeks if a fresh order is required.

MSS

Two reads to confirm the negative ICI finding. MMR IHC (MLH1, MSH2, MSH6, PMS2 on archival FFPE, one to two week turnaround) is the orthogonal protein-level confirmation on the NGS call. NGS-derived MSI calls have known false negatives at low tumor purity, and most ICI-trial central labs accept MMR IHC on its own. A confirmed MSS by both NGS and IHC closes the door cleanly on the tumor-agnostic pembrolizumab pathway.

The tumor microenvironment workup (PD-L1 22C3, CD3 / CD8 TIL density, alpha-SMA or H and E stromal density) frames any ICI-combination strategy. PDAC is the textbook immune-cold, stromally dense tumor, and the dominant interpretive question is usually T-cell exclusion rather than PD-L1 score. Returnable on a multiplex IHC panel (NeoGenomics MultiOmyx covers PD-L1, CD3, CD8, and stromal markers on a single section).

TMB

Documentation review rather than a new order. TMB values are sensitive to panel size, mutation-call filters, and whether synonymous variants are included; 4.1 mut/Mb on a 500-gene panel may map differently against the FDA-approved tumor-agnostic pembrolizumab threshold than a whole-exome-derived value (Friends of Cancer Research TMB Project harmonization study). If the existing report is from FoundationOne CDx, this is a same-day documentation check on the bioinformatics methods section. Below the 10 mut/Mb threshold the tumor-agnostic pembrolizumab route is foreclosed; the read confirms that foreclosure cleanly.

Where to order these assays

Assay	Provider	Decision gated	Contact
Orthogonal NGS panel plus ctDNA (KRAS G12R-specific ddPCR or comprehensive plasma NGS)	**Guardant Health (preferred) (Guardant360 CDx)**	**Pan-RAS / RAS(ON) inhibitor trials (RMC-6236 NCT05379985, RMC-7977 NCT06445062); G12C-specific agents are not the relevant class.**	**test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887**
Orthogonal NGS panel plus ctDNA	Foundation Medicine (FoundationOne Liquid CDx)	Pan-RAS / RAS(ON) inhibitor trials (RMC-6236 NCT05379985, RMC-7977 NCT06445062); G12C-specific agents are not the relevant class.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Orthogonal NGS panel plus ctDNA	Tempus (Tempus xF+ (plasma) / xT (tissue))	Pan-RAS / RAS(ON) inhibitor trials (RMC-6236 NCT05379985, RMC-7977 NCT06445062); G12C-specific agents are not the relevant class.	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Orthogonal NGS panel plus ctDNA	Natera (Signatera (tumor-informed MRD))	Pan-RAS / RAS(ON) inhibitor trials (RMC-6236 NCT05379985, RMC-7977 NCT06445062); G12C-specific agents are not the relevant class.	test info · 13011 McCallen Pass, Austin, TX 78753 · 1-650-249-9090
Orthogonal NGS panel plus ctDNA	Caris Life Sciences (Caris Assure)	Pan-RAS / RAS(ON) inhibitor trials (RMC-6236 NCT05379985, RMC-7977 NCT06445062); G12C-specific agents are not the relevant class.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
p16 IHC (clone E6H4 or equivalent)	**Mayo Clinic Laboratories (preferred)**	**CDK4/6-inhibitor eligibility (palbociclib, ribociclib, abemaciclib) and CDKN2A-loss-enriched trial arms.**	**test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710**
p16 IHC (clone E6H4 or equivalent)	ARUP Laboratories	CDK4/6-inhibitor eligibility (palbociclib, ribociclib, abemaciclib) and CDKN2A-loss-enriched trial arms.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787
p16 IHC (clone E6H4 or equivalent)	NeoGenomics Laboratories	CDK4/6-inhibitor eligibility (palbociclib, ribociclib, abemaciclib) and CDKN2A-loss-enriched trial arms.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
p16 IHC (clone E6H4 or equivalent)	LabCorp / Esoterix Oncology	CDK4/6-inhibitor eligibility (palbociclib, ribociclib, abemaciclib) and CDKN2A-loss-enriched trial arms.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
p16 IHC (clone E6H4 or equivalent)	Quest Diagnostics	CDK4/6-inhibitor eligibility (palbociclib, ribociclib, abemaciclib) and CDKN2A-loss-enriched trial arms.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378

CCND3 alteration-class refinement (amplification vs activating mutation vs fusion) via NGS plus FISH or copy-number array as needed	**Foundation Medicine (preferred) (FoundationOne CDx)**	**CDK4/6-inhibitor and cyclin-D-degrader trial enrichment (PF-07220060, fadraciclib, INCB123667).**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
CCND3 alteration-class refinement	Caris Life Sciences (Caris Molecular Intelligence)	CDK4/6-inhibitor and cyclin-D-degrader trial enrichment (PF-07220060, fadraciclib, INCB123667).	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
CCND3 alteration-class refinement	Tempus (Tempus xT)	CDK4/6-inhibitor and cyclin-D-degrader trial enrichment (PF-07220060, fadraciclib, INCB123667).	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
CCND3 alteration-class refinement	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	CDK4/6-inhibitor and cyclin-D-degrader trial enrichment (PF-07220060, fadraciclib, INCB123667).	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
CCND3 alteration-class refinement	NeoGenomics Laboratories (NeoTYPE Comprehensive Panel)	CDK4/6-inhibitor and cyclin-D-degrader trial enrichment (PF-07220060, fadraciclib, INCB123667).	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Comprehensive tumor NGS including SMAD4, KEAP1, STK11, GNAS, ARID1A	**Foundation Medicine (preferred) (FoundationOne CDx)**	**Frames prognosis under multi-agent therapy and weights any ICI-combo strategy on the KRAS axis.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
Comprehensive tumor NGS	Caris Life Sciences (Caris Molecular Intelligence)	Frames prognosis under multi-agent therapy and weights any ICI-combo strategy on the KRAS axis.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Comprehensive tumor NGS	Tempus (Tempus xT)	Frames prognosis under multi-agent therapy and weights any ICI-combo strategy on the KRAS axis.	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Comprehensive tumor NGS	NeoGenomics Laboratories (NeoTYPE Comprehensive Panel)	Frames prognosis under multi-agent therapy and weights any ICI-combo strategy on the KRAS axis.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Comprehensive tumor NGS	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	Frames prognosis under multi-agent therapy and weights any ICI-combo strategy on the KRAS axis.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Germline hereditary-cancer panel covering BRCA1, BRCA2, PALB2, ATM, MLH1, MSH2, MSH6, PMS2, EPCAM, CDKN2A, TP53, STK11	**Invitae (preferred) (Invitae Multi-Cancer Panel)**	**Olaparib maintenance after platinum for germline BRCA-mutant PDAC; family cascade testing; FAMMM-CDKN2A screening.**	**test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037**

Germline hereditary-cancer panel	GeneDx (GeneDx OncoGeneDx)	Olaparib maintenance after platinum for germline BRCA-mutant PDAC; family cascade testing; FAMMM-CDKN2A screening.	test info · 207 Perry Parkway, Gaithersburg, MD 20877 · 1-888-729-1206
Germline hereditary-cancer panel	Ambry Genetics (CancerNext)	Olaparib maintenance after platinum for germline BRCA-mutant PDAC; family cascade testing; FAMMM-CDKN2A screening.	test info · 1 Enterprise, Aliso Viejo, CA 92656 · 1-866-262-7943
Germline hereditary-cancer panel	Myriad Genetics (MyRisk Hereditary Cancer)	Olaparib maintenance after platinum for germline BRCA-mutant PDAC; family cascade testing; FAMMM-CDKN2A screening.	test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423
Germline hereditary-cancer panel	Color Health	Olaparib maintenance after platinum for germline BRCA-mutant PDAC; family cascade testing; FAMMM-CDKN2A screening.	test info · 831 Mitten Road, Burlingame, CA 94010 · 1-844-352-6567
MMR IHC panel (MLH1, MSH2, MSH6, PMS2)	**Mayo Clinic Laboratories (preferred)**	**Confirms the MSS call; closes the door on tumor-agnostic pembrolizumab.**	**test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710**
MMR IHC panel	ARUP Laboratories	Confirms the MSS call; closes the door on tumor-agnostic pembrolizumab.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787
MMR IHC panel	LabCorp / Esoterix Oncology	Confirms the MSS call; closes the door on tumor-agnostic pembrolizumab.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
MMR IHC panel	Quest Diagnostics	Confirms the MSS call; closes the door on tumor-agnostic pembrolizumab.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
MMR IHC panel	NeoGenomics Laboratories	Confirms the MSS call; closes the door on tumor-agnostic pembrolizumab.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
TMB assay platform and calling-pipeline confirmation (panel size, filters, harmonization with FDA-approved 10 mut/Mb threshold)	**Foundation Medicine (preferred) (FoundationOne CDx (TMB-harmonized))**	**Confirms sub-threshold TMB and the foreclosure of tumor-agnostic pembrolizumab eligibility.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
TMB assay platform confirmation	Caris Life Sciences (Caris Molecular Intelligence)	Confirms sub-threshold TMB and the foreclosure of tumor-agnostic pembrolizumab eligibility.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
TMB assay platform confirmation	Tempus (Tempus xT)	Confirms sub-threshold TMB and the foreclosure of tumor-agnostic pembrolizumab eligibility.	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137

TMB assay platform confirmation	NeoGenomics Laboratories (NeoTYPE Comprehensive Panel)	Confirms sub-threshold TMB and the foreclosure of tumor-agnostic pembrolizumab eligibility.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
TMB assay platform confirmation	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	Confirms sub-threshold TMB and the foreclosure of tumor-agnostic pembrolizumab eligibility.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
TP53 variant annotation (LOF vs dominant-negative vs gain-of-function) with reference to IARC TP53 database	**Memorial Sloan Kettering Diagnostic Molecular Pathology (preferred) (MSK-IMPACT (with variant annotation))**	**Frames any TP53-restoration or MDM2-inhibitor strategy; does not gate the KRAS axis.**	**test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000**
TP53 variant annotation	Foundation Medicine (FoundationOne CDx)	Frames any TP53-restoration or MDM2-inhibitor strategy; does not gate the KRAS axis.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
TP53 variant annotation	Tempus (Tempus xT)	Frames any TP53-restoration or MDM2-inhibitor strategy; does not gate the KRAS axis.	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
TP53 variant annotation	Caris Life Sciences (Caris Molecular Intelligence)	Frames any TP53-restoration or MDM2-inhibitor strategy; does not gate the KRAS axis.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
TP53 variant annotation	IARC TP53 Database (annotation reference)	Frames any TP53-restoration or MDM2-inhibitor strategy; does not gate the KRAS axis.	test info · 150 Cours Albert Thomas, 69372 Lyon CEDEX 08, France
p16 IHC across multiple tumor regions (primary site vs metastatic biopsy if available)	**Mayo Clinic Laboratories (preferred)**	**Refines confidence in the CDKN2A loss call for CDK4/6-inhibitor strategy; does not gate enrollment.**	**test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710**
p16 IHC multi-region	ARUP Laboratories	Refines confidence in the CDKN2A loss call for CDK4/6-inhibitor strategy; does not gate enrollment.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787
p16 IHC multi-region	NeoGenomics Laboratories	Refines confidence in the CDKN2A loss call for CDK4/6-inhibitor strategy; does not gate enrollment.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
p16 IHC multi-region	Memorial Sloan Kettering Diagnostic Molecular Pathology	Refines confidence in the CDKN2A loss call for CDK4/6-inhibitor strategy; does not gate enrollment.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
p16 IHC multi-region	LabCorp / Esoterix Oncology	Refines confidence in the CDKN2A loss call for CDK4/6-inhibitor strategy; does not gate enrollment.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
Serial KRAS G12R ctDNA quantitation (ddPCR or tumor-informed MRD) at baseline and at cycle 2 / cycle 4 of KRAS-directed therapy	**Natera (preferred) (Signatera (tumor-informed MRD))**	**Early-response and emerging-resistance surveillance on pan-RAS / RAS(ON) inhibitor therapy.**	**test info · 13011 McCallen Pass, Austin, TX 78753 · 1-650-249-9090**

Serial KRAS G12R ctDNA quantitation	Guardant Health (Guardant Reveal / Guardant360 Response)	Early-response and emerging-resistance surveillance on pan-RAS / RAS(ON) inhibitor therapy.	test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887
Serial KRAS G12R ctDNA quantitation	Bio-Rad / academic ddPCR core	Early-response and emerging-resistance surveillance on pan-RAS / RAS(ON) inhibitor therapy.	test info · 1000 Alfred Nobel Drive, Hercules, CA 94547 · 1-510-724-7000
Serial KRAS G12R ctDNA quantitation	Tempus (Tempus xM)	Early-response and emerging-resistance surveillance on pan-RAS / RAS(ON) inhibitor therapy.	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Serial KRAS G12R ctDNA quantitation	Foundation Medicine (FoundationOne Tracker)	Early-response and emerging-resistance surveillance on pan-RAS / RAS(ON) inhibitor therapy.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor NGS for KRAS copy-number (KRAS amplification, wildtype KRAS amplification, secondary KRAS mutations)	**Foundation Medicine (preferred) (FoundationOne CDx)**	**Frames durability and next-line planning for pan-RAS / RAS(ON) inhibitor therapy; does not gate enrollment.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
Tumor NGS for KRAS copy-number	Caris Life Sciences (Caris Molecular Intelligence)	Frames durability and next-line planning for pan-RAS / RAS(ON) inhibitor therapy; does not gate enrollment.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Tumor NGS for KRAS copy-number	Tempus (Tempus xT)	Frames durability and next-line planning for pan-RAS / RAS(ON) inhibitor therapy; does not gate enrollment.	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Tumor NGS for KRAS copy-number	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	Frames durability and next-line planning for pan-RAS / RAS(ON) inhibitor therapy; does not gate enrollment.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Tumor NGS for KRAS copy-number	NeoGenomics Laboratories (NeoTYPE Comprehensive Panel)	Frames durability and next-line planning for pan-RAS / RAS(ON) inhibitor therapy; does not gate enrollment.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Tumor microenvironment workup: PD-L1 IHC (22C3), CD3 / CD8 TIL density, and stromal density (alpha-SMA or H and E review)	**NeoGenomics Laboratories (preferred) (MultiOmyx multiplex IHC)**	**Frames ICI-combination strategy (KRAS-inhibitor plus PD-1, CXCR4 antagonist plus PD-1, vaccine plus PD-1); does not gate enrollment.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
Tumor microenvironment workup	Caris Life Sciences (Caris Molecular Intelligence)	Frames ICI-combination strategy (KRAS-inhibitor plus PD-1, CXCR4 antagonist plus PD-1, vaccine plus PD-1); does not gate enrollment.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669

Tumor microenvironment workup	Foundation Medicine (FoundationOne CDx (with PD-L1 IHC reflex))	Frames ICI-combination strategy (KRAS-inhibitor plus PD-1, CXCR4 antagonist plus PD-1, vaccine plus PD-1); does not gate enrollment.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor microenvironment workup	Mayo Clinic Laboratories	Frames ICI-combination strategy (KRAS-inhibitor plus PD-1, CXCR4 antagonist plus PD-1, vaccine plus PD-1); does not gate enrollment.	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
Tumor microenvironment workup	Memorial Sloan Kettering Diagnostic Molecular Pathology	Frames ICI-combination strategy (KRAS-inhibitor plus PD-1, CXCR4 antagonist plus PD-1, vaccine plus PD-1); does not gate enrollment.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
Orthogonal NGS panel plus ctDNA (KRAS G12R-specific ddPCR or comprehensive plasma NGS)	Pan-RAS and RAS(ON) multi-selective programs (RMC-6236, RMC-7977 and the divarasib-class follow-ons) require allele-resolved KRAS G12X status at trial entry, and most central labs re-test the variant on plasma at screening regardless of the local report. G12R is biologically distinct from G12C and G12D: it is so G12C-specific inhibitors (sotorasib, adagrasib) are not the relevant class. A second-platform call plus a baseline ctDNA reading anchors both eligibility and on-treatment monitoring.	Guardant Health (Guardant360 CDx)	10-20 mL Streck whole blood for ctDNA; archival FFPE or matched tissue NGS for the orthogonal call

p16 IHC (clone E6H4 or equivalent)	NGS copy-number calls for CDKN2A homozygous deletion can be unreliable at low tumor purity, and p16 IHC is the orthogonal protein-level readout that confirms loss of function. Most CDK4/6-inhibitor trials in CDKN2A-altered tumors require p16-negative status or biallelic CDKN2A loss at enrollment, and the IHC is the assay the central labs run. A confirmed loss reads cleanly; a discordance (NGS-loss but IHC-retained) is a meaningful signal that the call may be subclonal or assay-driven.	Mayo Clinic Laboratories	archival FFPE; same block as the tumor NGS
CCND3 alteration-class refinement (amplification vs activating mutation vs fusion) via NGS plus FISH or copy-number array as needed	The user-supplied report says CCND3 alteration without specifying class, and that distinction matters: an amplification or activating mutation supports cyclin-D-axis dependency and CDK4/6 sensitivity, whereas a single passenger missense without copy gain has weaker mechanistic weight. Cyclin-D-degrader and selective CDK4 programs (PF-07220060, fadraciclib, INCB123667) typically gate on copy-number gain or known activating mutations. Refine the class before the board ranks CDK4/6-inhibitor strategy alongside CDKN2A loss.	Foundation Medicine (FoundationOne CDx)	archival FFPE; usually returnable from the same NGS panel as the KRAS / co-mutation row

Comprehensive tumor NGS including SMAD4, KEAP1, STK11, GNAS, ARID1A	SMAD4 loss in PDAC is the strongest single co-alteration linked to wider metastatic spread and shorter survival, and it changes how the board weights aggressive multi-agent regimens against trial enrollment. KEAP1 and STK11 are imported from the NSCLC literature for a reason: they predict diminished ICI benefit on KRAS-mutant backgrounds and would tilt any decision to layer pembrolizumab onto a pan-RAS inhibitor. GNAS and ARID1A refine the differential against IPMN-origin disease and ARID1A-defined synthetic-lethal pathways.	Foundation Medicine (FoundationOne CDx)	archival FFPE; ctDNA as a backup if tissue is exhausted
Germline hereditary-cancer panel covering BRCA1, BRCA2, PALB2, ATM, MLH1, MSH2, MSH6, PMS2, EPCAM, CDKN2A, TP53, STK11	NCCN guidance is that every patient with pancreatic adenocarcinoma should be offered germline testing regardless of family history. A pathogenic BRCA1/2 or PALB2 variant opens the pathway (olaparib maintenance is FDA-approved for germline BRCA-mutant metastatic PDAC after platinum). Lynch and Li-Fraumeni hits would change family screening and would not be excluded by a somatic-only tumor panel; CDKN2A germline is non-trivial in PDAC with melanoma family history (FAMMM).	Invitae (Invitae Multi-Cancer Panel)	5-10 mL EDTA whole blood; saliva kits available from most providers

MMR IHC panel (MLH1, MSH2, MSH6, PMS2)	NGS-derived MSI calls have known false-negative cases at low tumor purity or when the panel uses fewer microsatellite loci. MMR IHC is the orthogonal protein-level confirmation and is the assay most ICI-trial central labs accept on its own. A confirmed MSS by both NGS and IHC closes the door cleanly on the tumor-agnostic pembrolizumab pathway and frees the board to focus the ICI discussion on combination strategies.	Mayo Clinic Laboratories	archival FFPE; same block as the tumor NGS
TMB assay platform and calling-pipeline confirmation (panel size, filters, harmonization with FDA-approved 10 mut/Mb threshold)	TMB values are sensitive to the panel size, the mutation-call filters, and whether synonymous variants are included. A 4.1 mut/Mb on a 500-gene panel may map differently against the FDA-approved tumor-agnostic pembrolizumab threshold than a whole-exome-derived value, and the harmonization study (Friends of Cancer Research TMB Project) shows non-trivial cross-platform spread. Confirming the platform and call pipeline locks in the sub-threshold call and forecloses the biomarker-agnostic pembrolizumab pathway cleanly. Below 10 mut/Mb, that route is off the table.	Foundation Medicine (FoundationOne CDx (TMB-harmonized))	no new tissue required

TP53 variant annotation (LOF vs dominant-negative vs gain-of-function) with reference to IARC TP53 database	The user report says inactivating mutation, but TP53 alterations split functionally into pure loss-of-function (truncations, splice), dominant-negative missense (hotspot R175H, R248W, R273H), and gain-of-function variants. Any TP53-restoration or mutant-p53 reactivation strategy (eprenetapopt / APR-246-class agents, MDM2 inhibitors in TP53-wildtype residual context) gates on the precise variant class. Annotate against IARC TP53 to make this explicit rather than re-litigating it at the trial-screening step.	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT (with variant annotation))	no new tissue required; re-read the existing NGS report
p16 IHC across multiple tumor regions (primary site vs metastatic biopsy if available)	PDAC biology often shows clonal heterogeneity between primary and metastatic sites, and CDKN2A loss is not always uniform. A patchy p16 result on multi-region IHC tempers expectations for CDK4/6-inhibitor monotherapy and may push the board toward combination strategies. Lower priority than the orthogonal IHC because the gating decision rides on the dominant block, but the heterogeneity read changes how the response is interpreted.	Mayo Clinic Laboratories	archival FFPE from a second site if available

Serial KRAS G12R ctDNA quantitation (ddPCR or tumor-informed MRD) at baseline and at cycle 2 / cycle 4 of KRAS-directed therapy	On KRAS-directed therapy, the rate of KRAS variant-allele-fraction decline by cycle 2 to cycle 4 is one of the earliest readouts of biological response and tracks ahead of RECIST imaging by weeks. A rising VAF on therapy is a sentinel for emerging resistance and prompts an earlier re-biopsy or re-staging conversation. Does not gate enrollment; informs how the team sequences imaging, re-biopsy, and switch decisions while on a pan-RAS inhibitor.	Natera (Signatera (tumor-informed MRD))	10 mL Streck whole blood per draw; serial sampling protocol
Tumor NGS for KRAS copy-number (KRAS amplification, wildtype KRAS amplification, secondary KRAS mutations)	Baseline KRAS amplification and acquired wildtype-KRAS amplification are documented resistance mechanisms to KRAS-directed therapy, including the pan-RAS class. Detecting a baseline amplicon shifts the durability expectation; detecting an emerging wildtype-allele amplification on progression reframes whether the next move is intra-class (different RAS-directed agent) or a switch to chemo or to a downstream pathway. The same NGS panel that returns the co-mutation profile should annotate these copy-number calls.	Foundation Medicine (FoundationOne CDx)	archival FFPE at baseline; ctDNA at progression

Tumor microenvironment workup: PD-L1 IHC (22C3), CD3 / CD8 TIL density, and stromal density (alpha-SMA or H and E review)	PDAC is the textbook immune-cold, stromally dense tumor, and the MSS / sub-threshold-TMB combination forecloses the easy ICI route. The microenvironment workup informs whether an ICI-combination strategy (CXCR4 antagonist plus PD-1, KRAS-inhibitor plus PD-1, vaccine plus PD-1) is biologically rational for this case or whether the dominant problem is exclusion of T cells from the tumor bed. A PD-L1 CPS read also feeds any ICI-combination trial that uses a soft enrichment threshold rather than a hard cutoff.	NeoGenomics Laboratories (MultiOmyx multiplex IHC)	archival FFPE; same block as the tumor NGS

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.